

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

Chapter 9: Some Applications of Trigonometry

Previous year board questions, topic-wise: 3 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

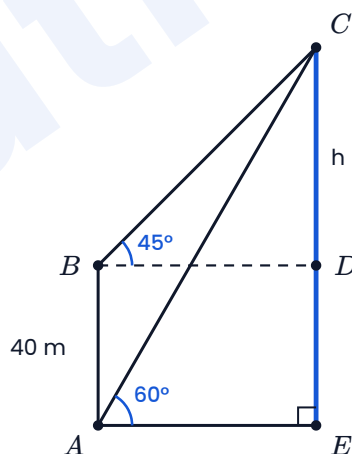
1 One right-angled triangle

1. A kite is flying 150 m above the ground. Its string is straight and makes an angle of 30° with the ground. How long is the string? **CBSE 2025, 1 mark**

2 Two right-angled triangles

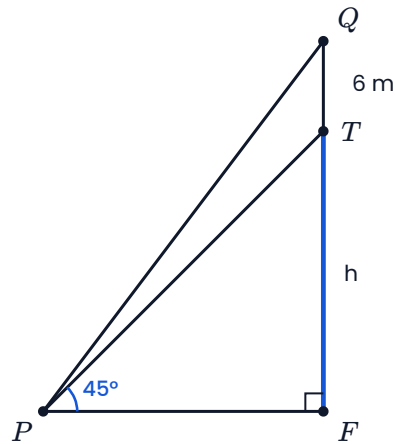
2. From a point A on the ground, the top C of a lighthouse CE is seen at an angle of elevation of 60° . From a deck at B , 40 m directly above A , the angle of elevation of C is 45° . D is the point on CE level with B , and $CD = h$ metres. (i) Write BD in terms of h . (ii) Write BC in terms of h . (iii) Find the height CE , taking $\sqrt{3} = 1.73$, OR find AE if $AC = 100$ m.

CBSE 2025, 4 marks



3. A 6 m pole stands on top of a tower. From a point P on the ground, the top of the pole is seen at an angle of elevation of 60° . From the top of the tower, P is seen at an angle of depression of 45° . Find the height of the tower and the distance of P from its foot. (Take $\sqrt{3} = 1.73$.)

CBSE 2024, 5 marks



FT is the tower and TQ is the pole

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